A New Yorkers' Guide to Buying 700 more cups of Coffee

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Github repo with commit

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1 Introduction

An average person living in New York spends 117 hours in traffic every year[1]. If we were to say that 1 hour of work is approximately $30 dollars in New York City, that is more than $3500 dollars lost per year. Or 700 coffees. In my analysis, my objective was to predict the median congestion time in a variety of suburbs scattered around NYC including the most congested ones. This study is aimed at the Government and Taxi companies to focus on these heavily congested locations and to employ strategies to mitigate congestion for civilians.

I chose between 2023 July and 2025 June to give me close to two years of training data to effectively make predictions on 2025 July. My data set also includes both green and yellow taxis, as i wanted the full scope of New York City.

1.1 External Dataset

I joined my Datasets on `date` with NYC weather data and preserved 2 columns which would be most valuable in my study.

- `temperature`: The temperature on that day.
- `visibility`: The visibility of that day.

I believe these 2 features perfectly capture the adversities faced by nature in a taxi journey that could increase congestion time. For example, a very warm day would be cause for more taxi journeys as people prefer to walk less or a snowy day increasing congestion time would have very low visibility.

1.2 Features of interest

I used the following features that originally existed in the dataset, to later create aggregated features that were a result of the group by on LocationID and date.

- `DOLocationID`: The LocationID of the taxi journey in my model.
- `trip_distance`: The distance in miles of the taxi journey.
- `fare_amount`: The fare amount of the taxi journey.
- `tip_amount`: The tip amount of the taxi journey.

- `total_amount`: The total amount charged to the passenger for the taxi journey.

Other features including PULocationID, date (as it was used to engineer new features) and several more were not used in my models as they did not seem to have any ability to predict congestion time from an intuitive standpoint.

1.3 Feature Engineering

In my project, i engineered the following features for my dataset:

- `travel_time_seconds`: The time taken in seconds for a taxi journey.
- `date`: The start date of the taxi journey.
- `congestion_time_seconds`: The estimated time a taxi spends in congestion in seconds. Assuming the average taxi speed in NYC is 14 mph [2], we convert this to miles per second by dividing by 3600. Congestion time is then computed as:

$$\texttt{congestion_time_seconds} = \texttt{travel_time_seconds} - \frac{\texttt{trip_distance}}{\texttt{average_speed}}.$$

  If the result exceeds a tolerance of 120 seconds (2 minutes), the result of the calculation is considered time spent in traffic or 0 otherwise.

1.3.1 Aggregated features

Later on in the project during model building a further set of features were built through aggregation to assist in forecasting a month ahead, using the initially engineered features. These include:

- `day_of_week`: The day of the week.
- `month`: The month the date belongs to.
- `median_congestion_time_seconds`: The median congestion time (seconds) for a passenger at a LocationID.
- `total_trips`: The total number of trips for a given LocationID (and later on also for a given date).
- `proportion_of_congestion_time`: The proportion of congestion time for a passenger at a LocationID.
- `several lag and rolling features (23 in total)`: These features were employed to assist in forecasting future congestion times by analyzing the past 4 days for each data point lag and 7 days into the past to compute a rolling mean.

2 Preprocessing

2.1 Data Filtering

There were several columns i filtered the dataset in, to have the most informative and accurate dataset. These include:

- `travel_time_seconds`: If the travel time was between 2 minutes (180 seconds) and 5 hours (18,000 seconds) i considered the taxi journey valid as a rational person would walk than rather

take a taxi for 2 minutes, and it did not sound realistic for a person to travel in a taxi for longer than 5 hours.

- `trip_distance`: The trip distance would have to be larger than a mile as no rational person is taking a taxi for a single mile.
- `fare_amount` and `total_amount`: The amount charged to the passenger would have to be larger than the initial fee of $3[3] to be considered a valid taxi journey.
- `congestion_time_seconds`: If the congestion time was 0 seconds i removed the taxi journey as these would prove non-informative to my analysis.

2.2 Dataset Shape

Dataset Stage	**Train Size**	**Test Size**
Unfiltered	57,506,050	24,386,791
Fully Filtered with Outlier Treatment	26,028,146	10,323,231
Grouped by LocationID and date	112,830	39,835
Grouped by LocationID and date for selected ID's	4,309	1,447

Table 1: Dataset Size throughout the project

However as the project approached the final modeling section the Train and Test Datasets were merged into one Dataset to forecast the unseen 2025 July data.

3 Analysis and Geospatial Visualization

3.1 2023 October and null taxi journeys

The 2023 October data, which in total spanned 3,588,462 journeys, were found mostly to have no pickup and drop off dates and times proving valueless to my study and were all dropped as they could not contribute to a temporal analysis. This brought the 2023 October data down to 66,088 valid taxi journeys.

3.2 Outlier Detection and Treatment

For a given dataset record if the `travel_time_seconds` and the `trip_distance` was outside the range of, IQR of that column multiplied by a factor:

$$\text{factor} = \sqrt{\ln(n)} - 0.5$$

where n is there size of the dataset, i considered it be an outlier. This factor is preferred over the usual 1.5, due to the magnitude of the Dataset. I chose these two columns specifically because they were most indicative of an invalid taxi journey.

3.3 Group by Data Analysis

Initially i wanted to explore what locations had the largest congestion times.

DOLocationID	median_congestion_time_seconds	total_trips
204	2151.5714285714284	1
105	952.8571428571428	9
156	866.5714285714284	3
46	837.1428571428573	57

Table 2: median congestion time at a location in descending order of `median_congestion_time_seconds` (Top 4)

But (Table 2) seemed unreliable as the highest congestion times had very few trips to that location and so, i instead found the proportion of congestion time relative to the trips to that location.

LocationID	median_congestion_time_seconds	total_trips	proportion_of_congestion_time
161	483.1428571428571	1,088,635	20.207594666393614
230	537.7142857142857	849,829	17.55657870192851
237	366.2857142857142	1,074,459	15.120515394593
236	328.1428571428571	1,094,226	13.795160285331116

Table 3: Proportion of median congestion time at a location in descending order of `proportion_of_congestion_time` (Top 4)

(Table 3) showed much more informative data, as now we can observe that most of Manhattan (Figure 1) was very highly congested, as was expected of it being the world's principal financial center[4].

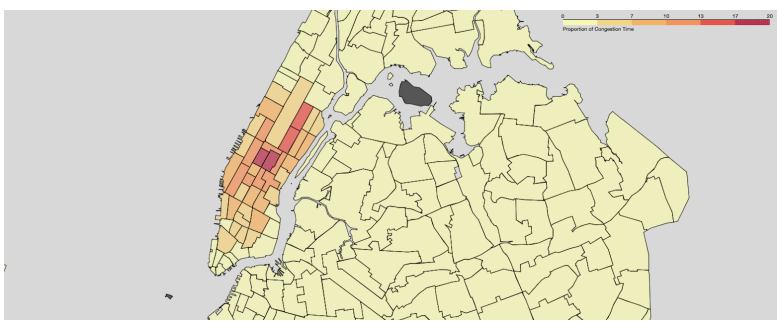

Figure 1: Proportion of median congestion time in New York City

3.4 Relationship between Congestion time and Travel time

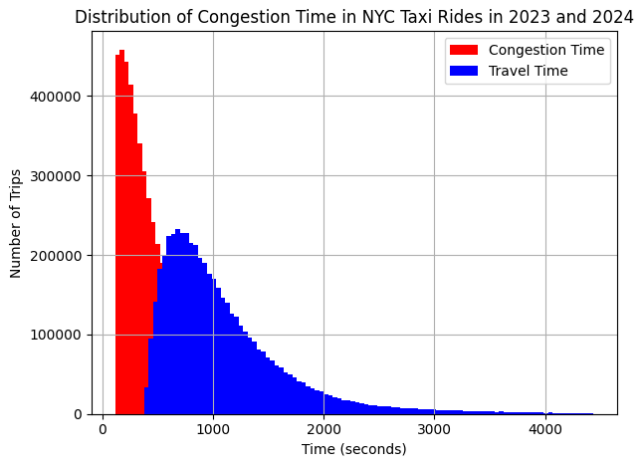

Figure 2: Distribution of Congestion time VS Travel time in New York City

As seen in (Figure 2) both of the data is heavily right skewed which lead to the initial calculation of median congestion time. From the distribution we see that a large proportion of the trips spent about 8 minutes in congestion (more than 450,000) which on our scale, is about a cup of coffee money. What can be also noticed is that, around 250,000 taxi journeys only traveled for around 12 minutes. Which means in theory, most passengers spend $\frac{3}{4}$th of their journey stuck in traffic.

3.5 Time Series Analysis

Using a collection of LocationID's with the 6 highest proportion of median congestion times along with two random LocationID's (123, 17) i plotted a Time Series of the last 3 months of median congestion time data.

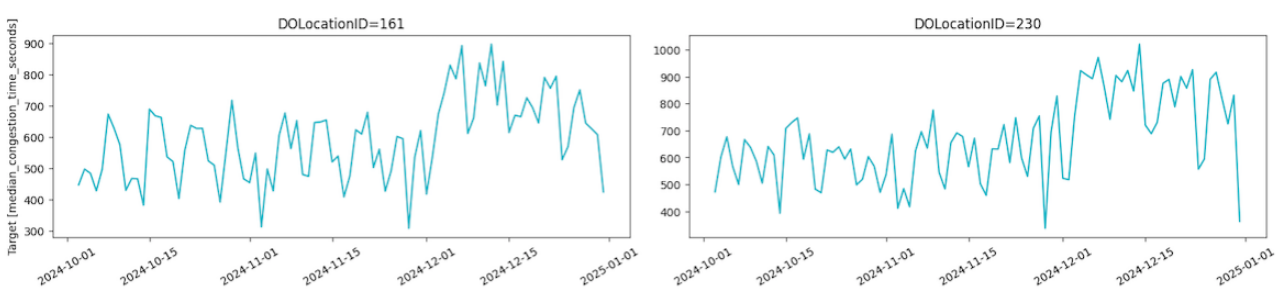

Figure 3: Time Series for last 3 months for Top 4 congested locations.

A certain cyclic pattern can be observed every 30 days in most locations as seen in (Figure 3) and this leads me to believe there is a strong correlation between each months median congestion time. Around the end of 2024 there can be seen a large decline in median congestion time which is synonymous of the Christmas and New Year holidays. This 30 day pattern led to the creation of 3 day lag when fitting the model, to capture some of the circular nature.

4 Modelling

All modelling is carried out on the 8 selected LocationID's.

4.1 Supervised Baseline Model: Lasso Regression

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_{\text{trip}}\,\text{trip_count}_i + \hat{\beta}_{\text{daily}}\,\text{daily_congestion}_i + \cdots + \sum_{k=1}^{K} \hat{\gamma}_k\,\mathbf{1}\{\text{DOLocationID}_i = k\} + \sum_{t=1}^{T} \hat{\delta}_t\,\mathbf{1}\{\text{date}_i = t\}.$$

Initially, to test the predictive power of the raw features (Original Dataset features) i built a linear regression model to predict the median congestion time. Due to the large sparsity created by the one hot encoded `DOLocationID` and `date` which i assumed had high predictive power, lasso was employed to reduce model complexity by reducing coefficients to 0 and performing feature selection. Further, all the input features were standardized to meet linear regression assumption. I used all of the 2023 and 2024 data to make predictions on 2025 and this supervised model performed quite reliably as seen in (Table 4).

Model	**R^2**	**RMSE**
Supervised Baseline Model	0.8544320568411927	41.565748729876475

Table 4: Baseline Model Metrics

As seen above in (Table 4) we can say that the model predicts 85% of the variance of the median congestion time and on average is 42 seconds off the actual time.

4.2 Semi-Supervised Models

Now to forecast the unseen and unlabeled 2025 July data, i first needed to check the validation of my models. To do this i split my Training data until 2025 February (about 80% of the combined Train and Test Datasets) and used the remaining data (2025 February to 2025 June) as my Validation Dataset. Using a combination of the aforementioned lag and rolling mean features and a couple of categorical variables, i devised 3 models to reverse engineer and predict the median congestion time alongside all the other raw features in the Dataset using multiple target prediction. The `DOLocationID`, `day_of_week` and `month` columns were one hot encoded due to the categorical nature and the input features were all standardized to meet the model assumptions (except in XGBRegressor model).

Model	**R^2**	**RMSE**
SGDRegressor	0.3351561521973351	87.77034232797322
XGBRegressor	0.3510047470201073	86.71789459177351
MLPRegressor	0.483297248979365	77.37637007174995

Table 5: Semi-Supervised Model Metrics

4.2.1 Model 1: Stochastic Gradient Descent Regressor (SGDRegressor) in combination with Multioutput Regression

Seeing the success of the linear regression model, i used a SGDRegressor as it is a scalable linear regression model that uses Stochastic Gradient Descent to optimize the linear objective function. The benefit it has in this context is due to the updating of model parameters using one sample at a time, the efficacy it brings about for larger datasets. Once again due to the sparsity of the one hot encodings

i opted for "elasticnet" penalty with a 75% lasso ratio to allow for the shrinking of coefficient to 0. According to (Table 5) around 34% of the variance of the median congestion time was accounted for by the model.

4.2.2 Model 2: XGBRegressor in combination with Multioutput Regression

I believed the features weren't linearly related to median congestion time, so I used a Tree-based model. XGBRegressor uses XGBoost, a gradient boosting framework that's efficient because it builds an ensemble of decision trees, correcting each other's errors. Tree models are robust to feature scaling and data types, so no preprocessing was needed. (Table 5) shows a slight improvement from the Model 1 but the improvement of less than 2% seems negligible. The improvement could be due to the large reduction in sparsity (removal of one hot encodings) from the nature of the model. Precautions for overfitting were taken in the hyperparameters (subsample, colsample_bytree) by subsampling on rows and features. L1 (lasso) and L2 (ridge) regularizations were also set in place.

4.2.3 Model 3:Multi-layer Perceptron regressor (MLPRegressor)

$$\hat{\mathbf{y}} = f\Big(\mathbf{W}_5\,\sigma\big(\mathbf{W}_4\,\sigma\big(\mathbf{W}_3\,\sigma\big(\mathbf{W}_2\,\sigma(\mathbf{W}_1\mathbf{x}+\mathbf{b}_1)+\mathbf{b}_2\big)+\mathbf{b}_3\big)+\mathbf{b}_4\big)+\mathbf{b}_5\Big)$$

(General Form for 4 hidden layers)

I finally used a feedforward neural network with four hidden layers and non-linear activation functions to capture complex, non-linear relationships between features and median congestion time. MLPRegressor also supports multi-output regression, allowing for shared representations across outputs. All target outputs were standardized to meet the assumption for a neural network. As shown in (Table 5), the model predicted about 50% of the variance in median congestion time and was on average off by 77 seconds. The neural network's ability to adapt to extensive feature vectors and approximate nonlinear relationships due to the complex interplay between features and targets contributed to this improvement.

LocationID 17 and 161 chosen to capture 2 extremes.

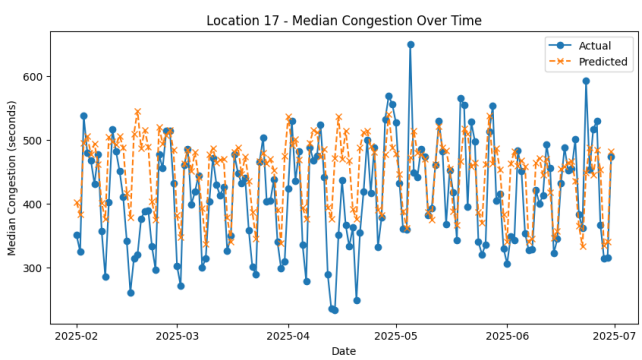

Figure 4: Predicted Vs Actual data for NN model for Location 17

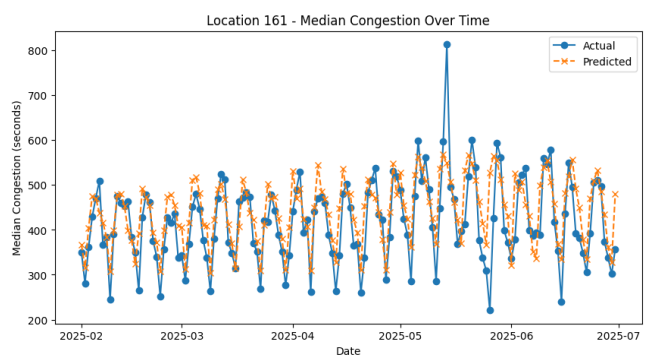

Figure 5: Predicted Vs Actual data for NN model for Location 161

In (Figure 4 and 5) we see that Model 3 generalizes well to the observed data in locations with very high congestion (Location 161) as they do not observe as much jumps in data points as to having many similar journeys, possibly caused by office commuters, however struggles to fit to the more niche locations (Location 17) with varying amounts of congestion that has no general pattern.

Even the best performing model only explaining half the variance of the target output is an invariant property of time data and the impermanent nature of temporal data.

4.3 Predicting 2025 July data

To predict the raw features for 2025 July data i utilized the most accurate Model 3.

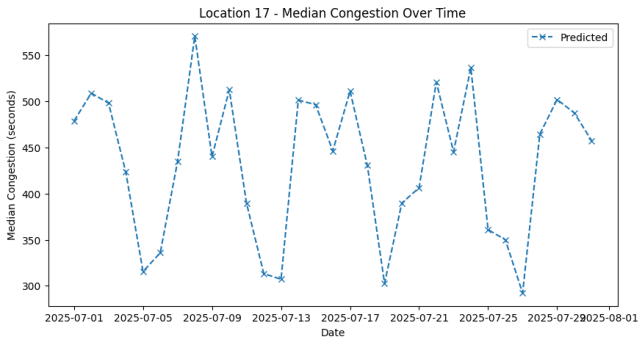

Figure 6: Forecast for July 2025: Location 17

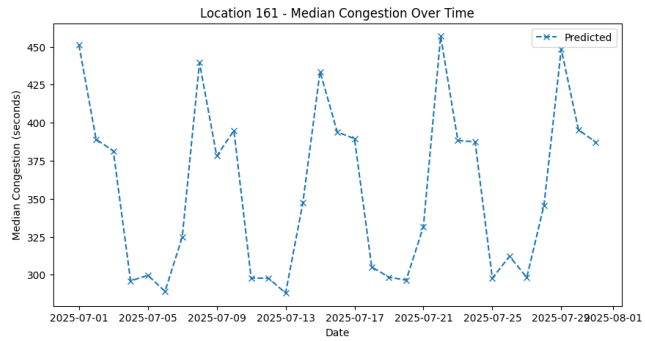

Figure 7: Forecast for July 2025: Location 161

The predictions made for location 161 (Figure 7) highlights on the weekly cyclic nature observed in (Figure 3) with a spike midweek and a steady decline in median congestion time towards the latter half of the week. Moreover, in locations like 17 (Figure 6) a spontaneous nature is seen. This could be a result of the high crime rate observed in the Brooklyn area and abrupt increases in road congestion due to road accidents and police chases. These randomly distributed median congestion over time upon further inspection could explore valuable insights for Governments.

5 Recommendations

In highly congested suburbs this peak is unavoidable but the government can take steps such as:

- Increasing number of lanes and road size.
- Improving traffic flow with extensive traffic light networks at the peak days of the week.
- Improve public transport so that civilians are incentivized to travel publicly, reducing taxi demand and increasing long term payoff for Governments.
- Improvement of road infrastructure and the Taxi company allocation to less serviced locations reducing overall taxi density in very congested areas.

6 Conclusion

The Model 3 outperformed in performance, but it struggled to predict travel times accurately in less congested areas. Dataset size may be a factor, but congestion time can only be partially modelled. The government and taxi companies can benefit from exploring this partially explained data.

To improve, build a more accurate model using advanced tools like Deep Learning libraries. Use other raw features generated by the models for more informative analysis and to counter randomness in less congested zones.

References

[1] Metropolitan Transportation Authority. *About Congestion Relief Zone*. https://congestionreliefzone.mta.info/about. 2025.

[2] Jiamin Han. *Looking Through the Taxi Meter: Analysis of the NYC Green Taxi Data*. https://medium.com/@jiaminhan/looking-through-the-taxi-meter-analysis-of-the-nyc-green-taxi-data-c1dbe5619afe. 2022.

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[4] Wikipedia. *Financial District, Manhattan*. https://en.wikipedia.org/wiki/Financial_District,_Manhattan. 2025.